EE3014/3042 ANSWER SHEET MIDTERM 2023-24 SPRING

**EE3014/3042 Midterm Exam** **Date: 16 April, 2024**

**Student Name and Number** Duration: 105 min.

**The notes and books will be kept closed. The calculator can be used. The necessary formulas are given on the last page.**

**Part1.**

1. **(25p)** A linear machine has magnetic flux density of 0.5 T directed into the page, a resistance of 0.25 Ω, a bar length of l = 1.0 m, and a battery of 100 V.
   - **(a)** What is the initial force on the bar at starting? What is the initial current flow?
   - **(b)** What is the no-load steady-state speed of the bar?
   - **(c)** If the bar is loaded with a force of 25 N opposite to the direction of motion, what is the new steady-state speed? What is the efficiency of the machine under these circumstances?
   - **(d)** Draw the plots for the velocity, induced force, induced voltage, and current as a function of time in the machine for parts (b), (c) and ~~(d)~~.
2. **(15p)** Suppose that the voltage applied to a load is V = 200∠-30° V and the current flowing through the load is I = 5∠15° A. Determine the complex power S consumed by the load, type of the load, and the power factor of the load. Also, draw the circuit and its phasor diagram.
3. **(25p)** A 100 - hp, 250 -V, 1200 r/min shunt dc motor with an armature resistance of 0.03 Ω and a field resistance of 41.67 Ω. Its magnetization curve is shown in Figure 1.The motor has compensating windings, so armature reaction can be ignored. Mechanical and core losses may be assumed to be negligible for the purposes of this problem. The motor is assumed to be driving a load with a line current of 126 A and an initial speed of 1103 r/min. To simplify the problem, assume that the amount of armature current drawn by the motor remains constant. What is the motor's speed if the field resistance is raised to 50 Ω?
4. **(25p)** A separately excited de generator is rated at 172 kW. 430 V, 400 A, and 1800 r/min. Its magnetization curve is shown in Figure 2. This machine has the following characteristics:

$R_f = 20\Omega$. $N_f = 1000$ turns. $R_A = 0{,}08\Omega$

   - **(a)** If the variable resistor $R_{adj}$ in this generator's field circuit is adjusted to 63 Ω and the generator's prime mover is driving it at 1600 r/min, what is this generator's no-load terminal voltage?
   - **(b)** What would its voltage be if a 360-A load were connected to its terminals? Assume that the generator has compensating windings.
   - **(c)** What would its voltage be if a 360-A load were connected to its terminals, but the generator does not have compensating windings? Assume that its armature reaction at this load is 450 A • turns.
   - **(d)** How much field current would be needed to restore the terminal voltage to its no-load value? (Assume that the machine has compensating windings.) What is the required value for the resistor $R_{adj}$ to accomplish this?

**Formulas**

| $P = E_A I_A$ | $Q = V_\phi I_A \sin\theta$ per phase | $P = V_\phi I_A \cos\theta$ per phase | $Pout = w\tau$ |
|---|---|---|---|
| Complex power S = VI* | $P = EI$, $F = i(lxB)$, $e_{ind} = (v \times B).l$ | $E_A = K\phi w$ | |
| | $P_{conv} = e_{ind}.i = F_{ind}.v$ | | |

**Part2(10p)** *Fill in the following blank(s),2 points each, and tick the correct answers( 1) and 2))*

2/ 1. If a conductor is wound around a piece of metal, such as iron or steel, and current is passed through that conductor, a magnetic field is produced around this assembly. It is called electromagnet

2/ 2. In a rotating field generator, direct current from a separate source is passed through commutator on the rotor by means of slip rings and brushes.

2/ 3. The armature windings are defined as the windings in which a voltage is induced, and the field windings are defined as the windings that produce the main magnetic flux in the machine.

2/ 4. The four equations that govern the behavior of a linear machine are 1. $\vec{F} = i(\vec{l} \times \vec{B})$ (induced force, Lorentz force), 2. $e_{ind} = (\vec{v} \times \vec{B}).\vec{l}$ (induced voltage (back emf)), 3. KVL (Kirchhoff's voltage Law), and 4. $F = ma$ (Newton's Law).

1/5

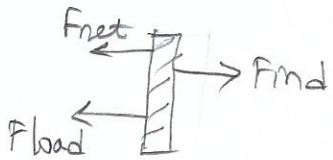
(25) 1) $B = 0.5\,⊙$, $R = 0{,}25\,\Omega$, $l = 1.0\,m$, $V_B = 100\,V$.

6/ a) $F_{init} = B i l = B \frac{V_B}{R} l = (0.5)\left(\frac{100}{0{,}25}\right)(1.0) = 200\,N$ where $i = 400\,A$

6/ b) $V_{ss,\,no\,load} = \frac{V_B = e_{ind}}{Bl} = \frac{100}{(0.5)(1.0)} = 200\,m/s.$

6/ c) If $F_{load} = 25\,N$, opposite direction to the motion

$V_{new,ss} = ?$




$f_{net} = F_{load} - F_{ind} = 0$ at steady-state

$F_{ind} = Bil$ ↑ until $F_{ind} = F_{load}$ at a lower speed.

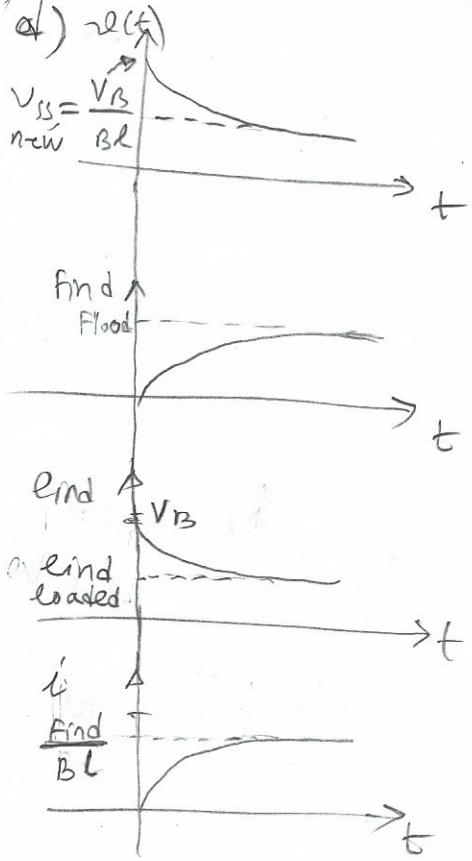
⇒ From $F_{ind} = Bil$ ⇒ $i = \frac{F_{ind}}{Bl} = \frac{25\,N}{(0.5)(1.0)} = 50\,A$ 1/

⇒ $e_{ind\,loaded} = V_B - iR = 100 - (50)(0{,}25) = 87.5\,V$ 1/

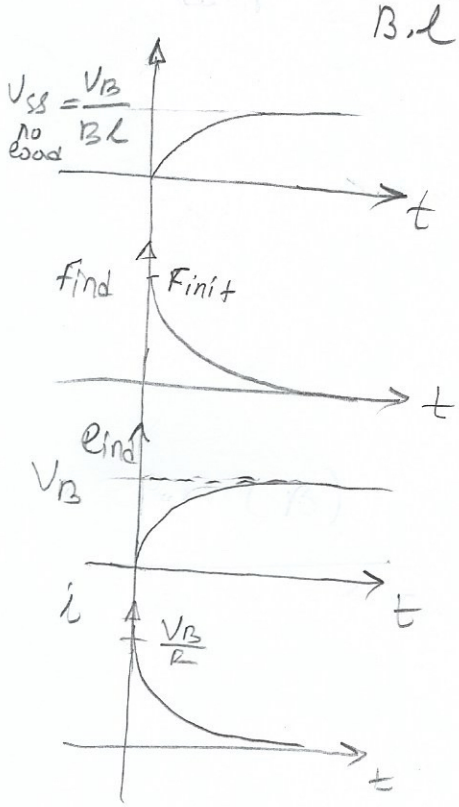
⇒ $e_{ind} = B v_{ss,new} l$ ⇒ $V_{ss,new} = \frac{e_{ind}}{B.l} = \frac{87.5}{(0.5)(1.0)} = 175\,m/s.$ 2/

7/

a)




Part c (loaded)




Part b (no load)

c) Efficiency

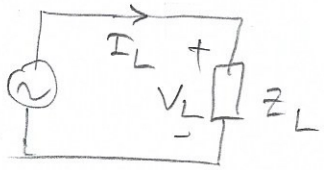
$P_{conv} = e_{ind} i = (87.5)(50) = F_{ind} v_{ss,new} = 4375\,W$

$P_{bat} = V_B i = (100)(50) = 5000\,W$

$\eta\% = \frac{P_{mech}}{P_{bat}} \times 100 = 87.5\%$ 2/

2/5

(15) 2)

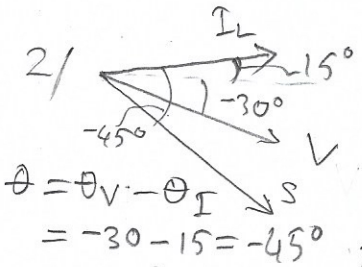



2/

$V = 200\angle -30°$ , $I_L = 5\angle 15° \Rightarrow S = ?$

Load type? PF?

$V = V_L$



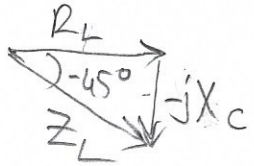

2/

$\theta = \theta_V - \theta_I$

$= -30 - 15 = -45°$ 1/

$$S = V I_L^* = 200\angle -30° \; 5\angle 15° = 1000\angle -15° \text{ VA}$$ 5/

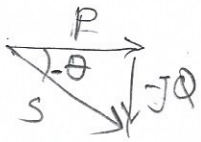
$$Z_L = \frac{V}{I_L} = \frac{200\angle -30°}{5\angle 15°} = 40\angle -45°$$ 2/




$Z_L = R_L - jX_C$ (capacitive load)

$PF = \cos(-45°) \cong 0{,}707$, leading 3/

$S = 1000\left[\cos(-45) + j\sin(-45)\right] = \underbrace{500\sqrt{2}}_{P} \underbrace{- j\,500\sqrt{2}}_{Q}$ VA




Since Q (reactive power) is negative (capacitive load), the PF is leading

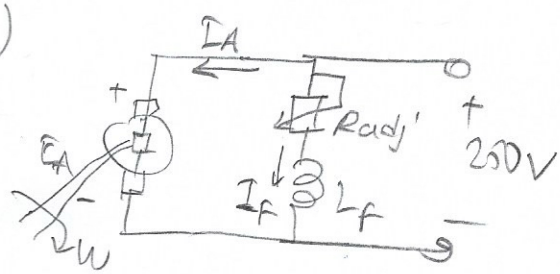
3) (25) 100hp, 250-V, 1200 r/min, shunt DC motor $\Rightarrow R_A = 0{,}03\,\Omega$ and $R_F = 41{,}67\,\Omega$

$I_L = 126$ A (line current), no mechanical loss, no AR

$n_{m1} = 1103$ r/min (initial speed)

$I_A$ is constant (assume)

$n_m = ?$ if $R_f = 50\,\Omega$ ↑




$I_{A1} = I_{L1} - I_{F1} = 126 - \frac{250}{41{,}67} = 120$ A (kept constant) assume 3/

$E_A = K'\Phi n_m \Rightarrow \frac{E_{A2}}{E_{A1}} = \frac{K'\Phi_2 n_{m2}}{K'\Phi_1 n_{m1}} = 1$ since $I_A$ is constant 2/

$E_{A1} = E_{A2} = E_A = V_T - I_{A1}R_A = 250 - (120)(0{,}03) = 246{,}4$ V 3/

$I_{f1} = \frac{V_T}{R_{F1}} = \frac{250}{41.67} \cong 6$ A $\Rightarrow n_{m1} = 1103$ r/min. 3/

↓ initial speed.

3/5

3) cnt. $R_{f2} = 50\ \Omega \Rightarrow I_{f2} = \frac{V_T}{R_{f2}} = \frac{250}{50} = 5A$, $\Rightarrow n_{m2} = ?$ 3/

The speed changes according to the magnetization curve (MC)

at $I_{f1} = 6A \Rightarrow E_A(I_{f1}) = 268V$ 3/

at $I_{f2} = 5A \Rightarrow E_A(I_{f2}) = 250V$ 3/ } from the MC.

$$\frac{\phi_2}{\phi_1}\frac{n_{m2}}{n_{m1}} = 1 \Rightarrow n_{m2} = \frac{\phi_1}{\phi_2}(n_{m1}) = \frac{268V}{250V}(1103\ r/min)$$

$$\Rightarrow n_{m2} = 1182.4\ r/min.$$ 5/

4) (25) A separately DC gen, rated at 172kW, 430V, 400A, 1800 r/min

$R_A = 0{,}08\ \Omega$, $R_F = 20\ \Omega$, $V_F = 430V$, $N_F = 1000$ turns/pole

a) $R_{adj} = 63\ \Omega$ (set), $n_{m1} = 1600$ r/min (prime mover driving speed)

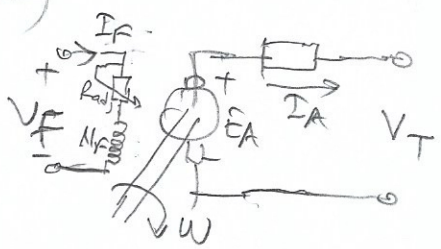
$V_{T,\text{no load}} = ?$

b) $V_T = ?$ if $I_L = 360$ with compensating windings

c) $V_T = ?$ if $I_L = 360$ without " "

d) $I_F = ?$ to restore $V_T$ to $V_{T,\text{no load}}$, with compensating windings and $R_{adj}$? (the required value)

a) $R_T = R_f + R_{adj} = 20 + 63 = 83\ \Omega$ 2/




$$I_F = \frac{V_F}{R_T} = \frac{430}{83} = 5{,}18A$$ 2/

From the MC $E_{A0} = 430V$ at $1800$ r/m. 2/

At 1600 r/min $\Rightarrow \frac{E_A}{E_{A0}} = \frac{k\phi n_m}{k\phi n_{m0}} \Rightarrow E_A = \frac{1600}{1800}(430) = 382.2V$ 2/

$V_{T,\text{no load}} = E_A = 382.2V$ at 1600 r/min. 2/

4/5

4-b) $I_L = 360A$ (the load connected), with compensating winding (cw)

$V_T = E_A - R_A I_A = 382.2 - (0,08)(360) = 353.4V$ 2/

c) $I_F^* = I_F - \frac{I_{AR}}{N_F} = 5.18 - \frac{450}{1000} = 4,73\ A$ 2/

From the MC $E_{A0} \cong 410V$ at 1800 r/min 2/

$E_A = \frac{1600}{1800}(410) = 364.4$ at 1600 r/min 2/

$V_T = E_A - R_A I_A = 364.4 - (0,08)(360) = 335.6\ V$ without cw. with the armature reaction. 2/

d) $E_A = V_{T,no\ load} + R_A I_A$ (to bring $V_T$ to $V_{T,no\ load}$ by varying $I_F$)

$= 382.2 + (0,08)(360) = 411V.$ 2/

The corresponding $E_A$ at 1800 r/min $= n_{m0}$

$\frac{E_A}{E_{A0}} = \frac{n_m}{n_{m0}} \Rightarrow E_{A0} = \frac{n_{m0}}{n_m} E_A = \left(\frac{1800}{1600}\right)(411) \cong 462.4V$ 1/

from the MC $I_F \cong 6.8A$ 1/

$I_F = \frac{V_F}{R_F + R_{adj}} \Rightarrow R_F + R_{adj} = \frac{V_F}{I_F}$

$R_{adj}' = \frac{V_F}{I_F} - R_F = \frac{430V}{6,8A} - 20\Omega = 43.24\Omega$ 1/

$R_{adj}'$: the required set value

5/5